Band-1 Computational Experiment: Agoh–Giuga Conjecture — Finite Giuga-Form Verification

Human + ZoraASI full-dash (executor)

2026-09-11

Band-1 Computational Experiment: Agoh–Giuga (Finite Verification)

Honesty stamp. This note **verifies** that for every integer n with $2 \le n \le N$, the Giuga congruence

$$\sum_{i=1}^{n-1} i^{n-1} \equiv -1 \pmod{n}$$

holds if and only if n is prime, for an explicit finite N computed on this box. It does **not** prove the **Agoh–Giuga / Giuga conjecture** for all n. Finite-range verification $\neq$ conjecture. **Status: OPEN.** This is **not** a Clay Millennium claim and **not** a ToE / GATE-04 unlock.

Abstract

We evaluate the Giuga sum form (preferred over modular Bernoulli) for $n = 2, \ldots, N$ and compare against a sieve primality oracle. We obtain

$$\mathbf{N = 20{,}000}, \qquad \mathbf{failures = 0}.$$

That is: every prime in the range satisfies the congruence, and every composite fails it (no Giuga-number counterexample in range). Intermediate certificate at $N = 5{,}000$: also **0** failures. Primary wall ≈ 72.6 s; total with units ≈ 77 s. **Conjecture status: OPEN.**

Problem statement

Giuga (1950). An integer $n \ge 2$ is prime if and only if $\sum_{i=1}^{n-1} i^{n-1} \equiv -1 \pmod{n}$.

Agoh form (equivalent). n is prime iff $n \cdot B_{n-1} \equiv -1 \pmod{n}$ (Bernoulli numbers). Equivalence: Agoh; Kellner (arXiv:math/0409259).

This experiment. Prefer the Giuga modular sum; verify equivalence to primality for all $n \in \{2, \ldots, N\}$ on this box; record any counterexample (expect none).

Literature ledger (not re-proved on box)

Year	Result	Authors	Status
1950	Giuga congruence as primality criterion	Giuga	conjecture stated

Year	Result	Authors	Status
1990s	Bernoulli form; equivalence	Agoh; Kellner	equivalence
1996	Counterexample $\geq\sim$ 13,800 digits	Borwein–Borwein–Borwein–Girgensohn	lit. bound
2013	Counterexample $\geq$ 4,771 distinct primes, $\geq$ 19,908 digits	Borwein–Maitland–Skerritt, *Integers* 13 #A67	published record
—	Structure: odd square-free Carmichael+Giuga	classical	proved constraints

Any counterexample would be a composite both Giuga and Carmichael; **none known**. Unconditional conjecture for all n remains **OPEN**.

Method (box)

1. **Unit tests.** Primes 2..23 hold; composites including 561 (smallest Carmichael) fail; full equiv on 2..200.
2. **Sieve.** Boolean primality table to N.
3. **Giuga sum.** For each n, compute $\sum_{i=1}^{n-1} \mathrm{pow}(i, n-1, n) \bmod n$ and test equality to $n-1$.
4. **Failure log.** Primes missing the congruence, or composites holding it (expect **none**).

Not used: modular Bernoulli B_{n-1} (equivalent but harder); literature digit bounds (ledger only).

Results

Quantity	Value
Primary N	**20,000**
n checked	19,999
Primes / composites	2,262 / 17,737
fails_prime	[]
counterexamples (composite holds)	[]
failures_total	**0**
Intermediate $N = 5{,}000$ failures	0 (wall ≈ 4.4 s)
Unit tests	PASS
Primary wall	≈ 72.57 s

Equivalence holds throughout the box range. No counterexample recorded.

Honesty one-liner

Finite only ($N = 20000$); Agoh–Giuga conjecture remains OPEN; no Clay/ToE.

Artifacts

- Code: `/workspace/experiments/band1/agoh_giuga/`
- `results.json`, `literature_ledger.json`, `SUMMARY.txt`
- Log: `/workspace/experiments/logs/2026-09-11_agoh_giuga.log`